Guojun Xu

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Research Profile

Ph.D. student in Computer Science and Technology at Wuhan University of Technology, advised by Prof. Junwei Zhou. My research focuses on distributed image compression, multimodal fusion, large-model-based image compression, and the perception–distortion trade-off.

Education

Wuhan University of Technology Sep. 2024 – Present

Direct-entry Ph.D. in Computer Science and Technology
Advisor: Junwei Zhou

Wuhan University of Technology Sep. 2020 – Jun. 2024

B.S. in Information and Computing Science

Publications

[1] **Guojun Xu**, Mingyang Zhang, Jianwen Xiang, Cheng Tan, Yanchao Yang, and Junwei Zhou. "Distributed Image Compression with Multimodal Side Information at Extremely Low Bitrates." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. **Accepted.**

[2] **Guojun Xu**, Cheng Tan, Jianwen Xiang, and Junwei Zhou. "CODIC: Distributed Image Compression with Distortion–Perception Co-Optimization." *IEEE Transactions on Multimedia*, 2026. doi:10.1109/TMM.2026.3718254.

Manuscripts Under Review

[3] **Guojun Xu** et al. "Perceptual Distributed Image Compression via Controlled Stochastic Reconstruction." *Data Compression Conference (DCC)*, 2027. **Under Review.**

[4] **Guojun Xu** et al. "Conditional Residual Decomposition for Distributed Image Compression." *International Conference on Learning Representations (ICLR)*, 2027. **Under Review.**

[5] **Guojun Xu** et al. "Retrieval-Augmented Distributed Image Compression with Multi-Reference Side Information." *IEEE Transactions on Image Processing (TIP)*. **Under Review.**

[6] **Guojun Xu** et al. "Distributed Image Compression with Target-Consistent Multimodal Side Information at Extremely Low Bitrates." *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*. **Under Review.**

Research Contributions

Perception–distortion trade-offs [2, 3]

Derived a local Gaussian analysis linking reconstruction scale and source–reconstruction correlation to the distortion cost of stochastic refinement (SPDIC). Developed distortion-oriented and diffusion-based branches with adaptive regional fusion (CODIC).

Multimodal conditioning and side-information reliability [1, 6]

Developed a diffusion decoder combining text derived from correlated images with visual side features. A multimodal alignment objective supervises feature masks for semantic and fine-detail recovery. Extended this direction with reliability-aware conditioning based on bidirectional latent matching and state-aware gating.

Conditional residuals and reconstruction uncertainty [4]

Applied conditional-expectation projection to separate residuals into side-information-predictable and orthogonal components, guiding deterministic correction and stochastic uncertainty modeling.

Retrieval-based multi-reference compression [5]

Retrieved references from a decoder-side gallery using the transmitted representation, then combined depth-aware geometric alignment with multi-reference cross-attention.

Prospective Research Directions

Distributed image compression; multimodal fusion; large-model-based image compression; perception–distortion trade-off theory.

Projects

Selected Participant, CAST Special Talent Cultivation Project for PhD Students 2026

Cultivating Society: China Society of Image and Graphics (CSIG)

Graduate Student Free Exploration Project, WHUT Independent Innovation Research Fund 2025

Wuhan University of Technology

Scholarships

First-class Academic Scholarship for Ph.D. Students 2026

Wuhan University of Technology

Second-class Academic Scholarship for Ph.D. Students 2025

Wuhan University of Technology

First-class Academic Scholarship for Ph.D. Students 2024

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